

Member ID: _____

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LINUX OPERATING SYSTEM FUNDAMENTALS (350)

NATIONAL 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice: Identify the letter of the choice that *best* completes the statement or answers the question.

1. Redhat Enterprise Linux (RHEL) changed its default firewall starting in versions 7.0 and newer. What did they change to?
 - a. Iptables
 - b. Firewallld
 - c. ufw
 - d. IPSEC
2. When using the FirewallD firewall what command below would open up port 443?
 - a. firewall-cmd --add-port
 - b. firewall-ctl --add-port=443/tcp
 - c. firewall-cmd --add-port=443/tcp OR firewall-cmd --add-service=https
 - d. firewallld --add-service=https
3. The command scp can be used to transfer files to other servers using ssh protocol.
 - a. True
 - b. False
4. What option can you use with scp to copy an entire directory?
 - a. -R
 - b. -r
 - c. -S
 - d. -s
5. What command below would copy the file named info.txt from your home folder to the /Shared folder on another server called ServerB using the admin user account?
 - a. scp admin@ServerB:/Shared ~/info.txt
 - b. scp ~/info.txt admin@ServerB:/Shared
 - c. scp ~/info.txt ServerB:admin/Shared
 - d. scp ServerB:admin/Shared ~/info.txt
6. This server will no longer be sharing files via the samba service and you need to block the service port in Ubuntu using the ufw firewall. Which command(s) below will accomplish this?
 - a. sudo ufw deny smb
 - b. sudo ufw block samba
 - c. sudo ufw block smb
 - d. sudo ufw deny samba

7. A server called ServerA is exporting an NFS share from the /Backup directory, which command would properly mount that share on your Ubuntu system in the /mnt/backup folder?
 - a. `sudo mount -t nfs //ServerA/Backup /mnt/backup`
 - b. `sudo mount -t nfs ServerA:/Backup /mnt/backup`
 - c. `sudo mount -t nfs /Backup@ServerA /mnt/backup`
 - d. `sudo mount -t nfs //ServerA:/Backup /mnt/backup`
8. Which fstab entry listed below would correctly mount an unprotected samba share on ServerB called Shared to the local directory /mnt/Shared?
 - a. `//ServerB/Shared /mnt/Shared cifs guest,uid=1000,icharset=utf8 0 0`
 - b. `//Shared@ServerB /mnt/Shared cifs guest,uid=1000,icharset=utf8 0 0`
 - c. `ServerB:/Shared /mnt/Shared cifs guest,uid=1000,icharset=utf8 0 0`
 - d. `Shared@ServerB /mnt/Shared guest,uid=1000,icharset=utf8 0 0`
9. Which file is used to enable and configure nfs shares on a Linux server?
 - a. `/etc/nfsmount`
 - b. `/etc/exports`
 - c. `/etc/nfs`
 - d. `/etc/exportfs`
10. Which fstab entry listed below would correctly mount an NFS share on ServerC exported from /Backup to the local directory /mnt/backup on your Ubuntu system?
 - a. `//ServerC/Backup /mnt/backup nfs defaults 0 0`
 - b. `ServerC:/Backup /mnt/backup nfs defaults 0 0`
 - c. `//ServerC:/Backup /mnt/backup nfs defaults 0 0`
 - d. `/Backup@ServerC /mnt/backup nfs defaults 0 0`
11. Which command would display all messages from the kernel ring buffer a Linux system?
 - a. `journalctl`
 - b. `dmesg`
 - c. `mesg`
 - d. `kmesg`
12. On a server with Redhat Linux installed, which log file can be used to show info on failed logins?
 - a. `/var/log/access`
 - b. `/var/log/login`
 - c. `/var/log/secure`
 - d. `/var/log/messages`
13. Which command will display information about PCI devices in Linux?
 - a. `lsmod`
 - b. `shpci`
 - c. `showpci`
 - d. `lspci`

14. Which command would show you the driver and its version used for the NIC named eth0 on a Linux server?
 - a. ip show eth0
 - b. modinfo eth0
 - c. ethtool -i eth0
 - d. ifconfig eth0
15. What command can be used to display version info for the storage driver named megasas in Linux?
 - a. info megasas
 - b. modinfo megasas
 - c. lsmod megasas
 - d. version megasas
16. Which Linux tool can be used to create and manage software raid devices?
 - a. mdadm
 - b. dmraid
 - c. modraid
 - d. rdadm
17. On an Ubuntu system which command will display the required packages to install vim?
 - a. sudo apt list vim
 - b. sudo apt depends vim
 - c. sudo apt cache vim
 - d. sudo apt show vim
18. Which command on a Linux NFS Server would show all clients with NFS shares mounted?
 - a. nfsmount
 - b. showmount
 - c. nfsstat
 - d. shownfs
19. Which steps below would be the correct order to expand LVM Storage?
 - a. Format disk/partition with pvcreate, extend logical volume, extend volume group, resize partition
 - b. Extend volume group, format disk/partition with pvcreate, resize partition, extend logical volume
 - c. Format disk/partition with pvcreate, extend volume group, extend logical volume, resize partition
 - d. Extend volume group, format disk/partition with pvcreate, extend logical volume, resize partition

20. What command below would add a new disk, sdc1, to an existing volume group named Data?

- a. vgextend Data /dev/sdc1
- b. lvextend /dev/sdc1 Data
- c. vgextend /dev/sdc1 Data
- d. lvextend Data /dev/sdc1

21. The XFS file system can be expanded and shrunk as needed.

- a. True
- b. False

22. It is possible to create a software RAID using logical volumes created with LVM.

- a. True
- b. False

23. Looking at the below output of the lvs command what is the name of the volume group on this Linux server?

```
root centos -wi-ao---- 49.00g
swap centos -wi-ao---- 6.00g
```

- a. root
- b. swap
- c. centos
- d. None of these are correct

24. Which command was used to display the output shown below?

PV	VG	Fmt	Attr	PSize	PFree
/dev/nvme0n1p1	Datastore	lvm2	a--	<223.57g	80.00m
/dev/sda1	Datastore	lvm2	a--	<3.64t	88.00m
/dev/sdb1	Datastore	lvm2	a--	<3.64t	0
/dev/sdc2	rhel	lvm2	a--	<54.78g	192.00m

- a. blkid
- b. vgs
- c. pvs
- d. lsblk

25. Which command will show the health of a Linux software RAID disk labeled as “md0”?

- a. cat /proc/mdstat OR mdadm --detail /dev/md0
- b. mdadm --manage /dev/md0
- c. mdadm --detail /dev
- d. mdadm --status /dev/md0

26. Which command will display the output seen below?

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINT
sda	8:0	0	24G	0	disk	
└sda1	8:1	0	24G	0	part	/
sdb	8:16	0	16G	0	disk	
└md0	9:0	0	16G	0	raid1	
sdc	8:32	0	16G	0	disk	
└md0	9:0	0	16G	0	raid1	
sr0	11:0	1	1024M	0	rom	

- a. blkid
- b. lsblk
- c. df
- d. fdisk

27. A Linux server called ServerA has just been setup on the network with the SSH service installed and running using default settings but when you try to connect via ssh you receive the error below, what would be the most likely cause?

ssh: connect to host ServerA port 22: Connection refused

- a. Connecting to the wrong port for SSH
- b. No network connection to ServerA
- c. None of these are correct
- d. Firewall on ServerA blocking port 22

28. Which would *not* be a valid reason for the error seen below?

root@ubuntu01:~# mount -t nfs ServerA:/Data /mnt/nfs

mount.nfs: access denied by server while mounting ServerA:/Data

- a. No entry in NFS exports file for /Data
- b. No NFS server running on ServerA
- c. Your system you are connecting from is in the Servers hosts.deny file
- d. Your system's hostname/IP is not listed in ServerA's exports file for /Data

29. What Linux command would allow you to view all set permissions and ACL's on a file/folder?

- a. ls -la
- b. showacl
- c. getfacl
- d. chmod

30. What Linux command would you use when you need to set multiple permissions for different users and groups on specific folders/files?

- a. chown
- b. chmod
- c. setfacl
- d. usermod

31. Which option would you use with the Linux command setfacl to remove either user or group permissions from a file/folder?

- a. -d
- b. -x
- c. -m
- d. -r

```
# file: Marketing/
# owner: root
# group: root
user::rwx
user:tom:---
user:larry:r-x
user:susan:rwx
group:---
group:admins:rwx
mask::rwx
other:---
```

Image 3A: In the Image above you can see the ACL permissions for the folder Marketing on our Linux system. The group admins contains both users tom and larry.

32. Referring to the ACL's seen in **Image 3A** above which users would have access to view files in the Marketing folder?

- a. Susan and Larry
- b. Susan
- c. Tom and Larry
- d. Susan, Tom and Larry

33. Referring to the above **Image 3A** again which users would have permission to access and write to files in the Marketing folder?

- a. Susan
- b. Susan and Larry
- c. Susan, Tom, and Larry
- d. Tom and Larry

34. Which command would give the group Sales read/write/execute permissions to all current and newly created files/folders in the Sales folder?

- a. setfacl -m u:Sales:rwx Sales
- b. setfacl -a d:g:Sales:rwx Sales
- c. setfacl -a g:sales:rwx Sales
- d. setfacl -m d:g:Sales:rwx Sales/

35. Which option used with setfacl command would remove all user/group permissions set on a file/folder?
- d
 - b
 - x
 - r
36. A user or group only needs read permissions set to view contents of a directory.
- True
 - False
37. Which nmcli command below would properly change the Network adapter ens192 from dhcp to static setting?
- sudo nmcli connection modify ens192 ipv4.method static
 - sudo nmcli device modify ens192 ipv4.method static
 - sudo nmcli device modify ens192 ipv4.method manual
 - sudo nmcli connection modify ens192 ipv4.method manual
38. On Ubuntu versions below 20.04 what file could you manually edit to configure the network adapters without using network manager tools?
- /etc/sysconfig/network-scripts/ifcfg
 - /etc/network/interfaces
 - /etc/sysconfig/network/interfaces
 - /etc/network/ifcfg
39. Which Linux command would list each network adapter and their current link state?
- ip state
 - ip link
 - ip show
 - ip list
40. Which nmcli command below would configure the network adapter ens192 with ip/subnet of 192.168.1.2/255.255.255.0?
- sudo nmcli connection modify ens192 ipv4.addresses 192.168.1.2/255.255.255.0
 - sudo nmcli device modify ens192 ipv4.addresses 192.168.1.2/255.255.255.0
 - sudo nmcli connection modify ens192 ipv4.addresses 192.168.1.2/24
 - sudo nmcli device modify ens192 ipv4.addresses 192.168.1.2/24
41. In Redhat Linux what file can you manually edit to configure a network adapter named em1 without using network manager?
- /etc/sysconfig/network-scripts/ifcfg-em1
 - /etc/network/interfaces
 - /etc/sysconfig/network/interfaces
 - /etc/network/ifcfg-em1

```
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: inactive (dead) since Wed 2021-12-29 22:45:51 CST; 1s ago
     Process: 6084 ExecStart=/usr/sbin/sshd -D $SSH_OPTS (code=exited, status=0/SUCCESS)
    Main PID: 6084 (code=exited, status=0/SUCCESS)
```

Image 4A: Showing the status of sshd service

42. Looking at Image 4A above you can confirm the ssh service is currently running.
- True
 - False
43. Looking at the Image 4A above you can confirm the ssh service is set to start automatically at boot.
- True
 - False
44. Which Linux command can be used to remove a user?
- usermod
 - userdel
 - removeuser
 - remuser
45. Which command below would add a new user named John and add it to the group to Sales?
- sudo usermod John -g Sales
 - sudo useradd -G Sales John
 - sudo useradd -G John Sales
 - sudo usermod -g Sales John
46. Which command would add the user named Susan to the group Marketing without changing its default group?
- sudo usermod -G Marketing Susan
 - sudo usermod -g Marketing Susan
 - sudo usermod -aG Marketing Susan
 - sudo usermod -ag Marketing Susan
47. In modern Redhat Linux OS's you can use both yum and dnf to update packages.
- True
 - False

48. In Ubuntu if you want to update all packages including kernel packages you would run: “sudo apt dist-upgrade”.
- a. True
 - b. False
49. In Ubuntu, which command would only update package repositories and package lists available?
- a. sudo apt upgrade
 - b. sudo apt udpate
 - c. sudo apt refresh
 - d. sudo apt dist-update
50. In Ubuntu, which command would search repositories for keyword http?
- a. sudo apt repo search http
 - b. sudo apt list http
 - c. sudo apt search http
 - d. sudo apt find http